Coastal Vulnerability and 3D Planning Visualisation - Ayetoro, Nigeria

Building-level exposure screening for climate-adaptation planning

Abdullah Abdazeez Ayomide

Executive Summary

Ayetoro is a low-lying coastal settlement in Ilaje LGA, Ondo State. This project assesses relative building-level coastal vulnerability using shoreline proximity, low elevation, building density, road density and facility exposure. The final validated inventory contains 1,628 buildings, and the resulting Coastal Vulnerability Index (CVI) shows a strong concentration of exposure: 1,175 buildings (72.17%) fall in the High or Very High classes.

The continuous CVI ranges from 0.4142 to 0.7500, with a mean of 0.6421. Validation against an independent historical coastal-impact band produced 92.74% recall, 93.45% precision, 82.66% specificity and a continuous CVI ROC-AUC of 0.9531.

1. Planning Question

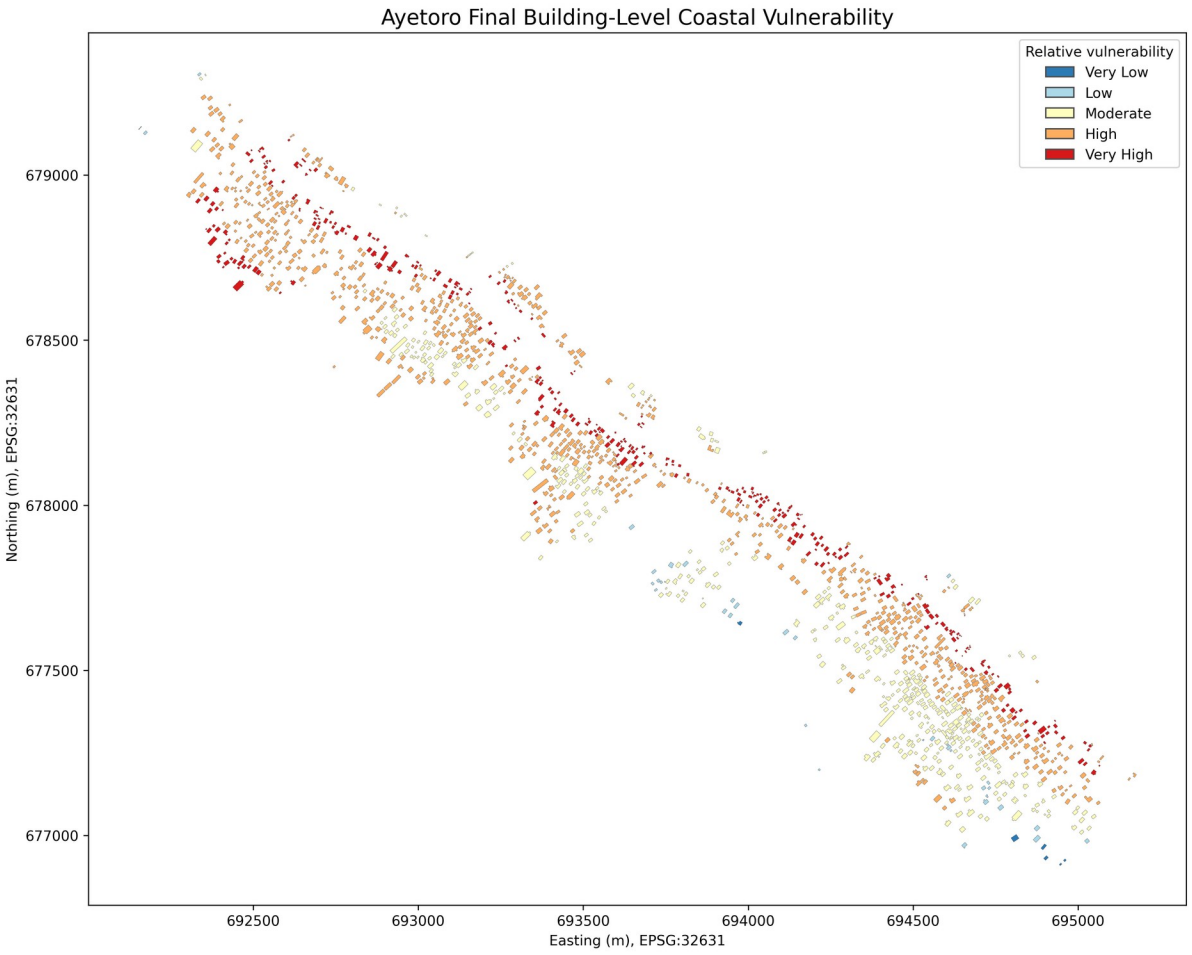
Which buildings in Ayetoro are relatively more exposed to coastal pressure, and how can that information support adaptation planning?

2. Building Inventory

The final building inventory contains 1,463 OpenStreetMap footprints retained after review and 165 conservative Google Open Buildings V3 supplements, producing a total of 1,628 buildings. The supplementary detections increased the refined inventory by 11.28%.

3. Coastal Vulnerability Model

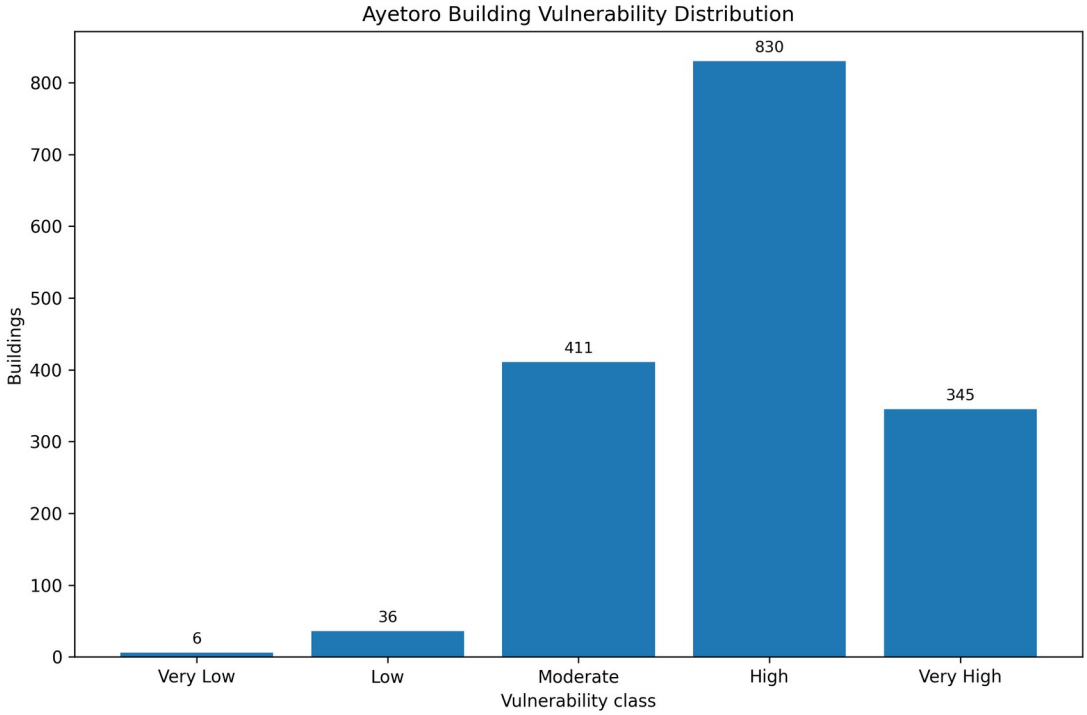
The final CVI combines five indicators: shoreline proximity (30%), low elevation (25%), building density (20%), road density (15%) and facility exposure (10%). Building height is not part of the CVI.



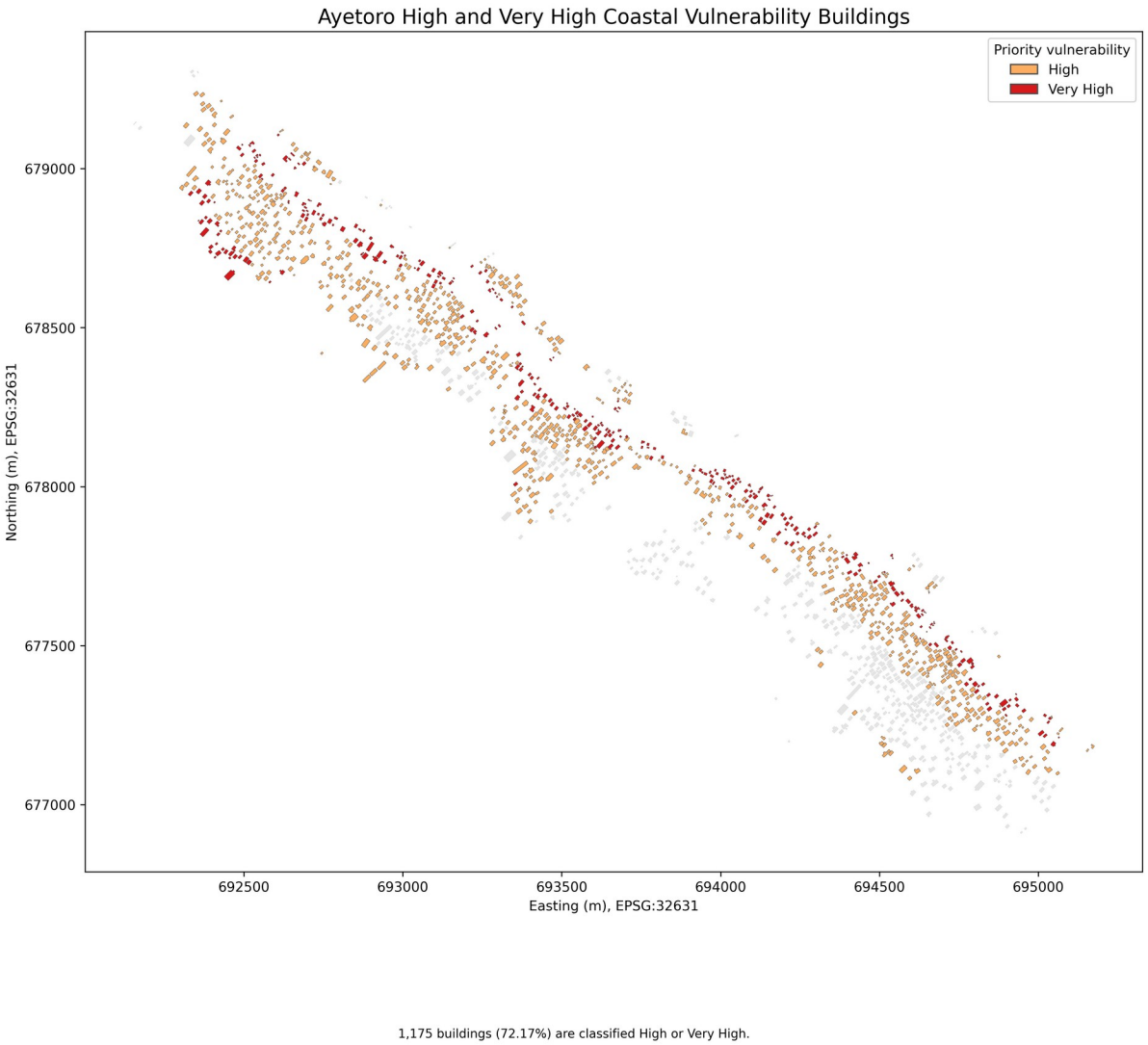
Final building-level coastal vulnerability in Ayetoro.

4. Vulnerability Distribution

The final classification contains 6 Very Low buildings (0.37%), 36 Low (2.21%), 411 Moderate (25.25%), 830 High (50.98%) and 345 Very High (21.19%). High and Very High classes together contain 1,175 buildings, equivalent to 72.17% of the validated inventory.



Final distribution of building vulnerability classes.

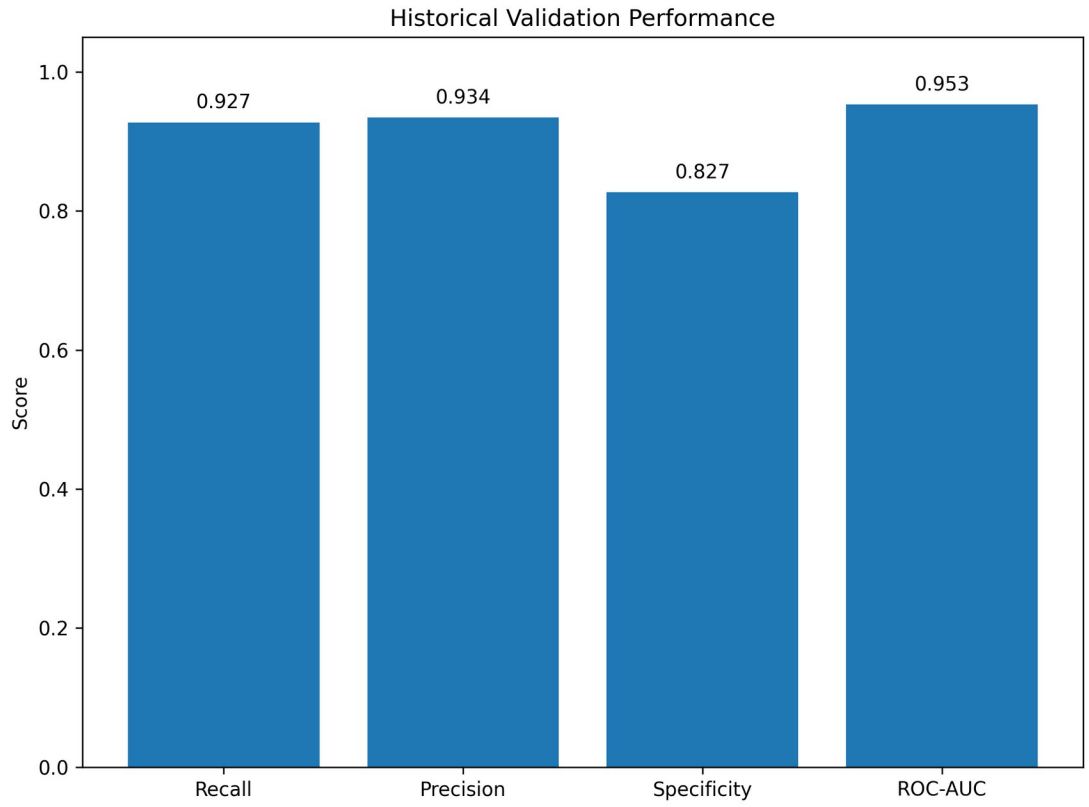


High and Very High vulnerability focus areas.

5. Historical Validation

Historical evidence records shoreline encroachment of about 145.62 m between March 2018 and February 2023, together with documented land and building losses. This evidence was used only as an independent validation reference rather than as a model input.

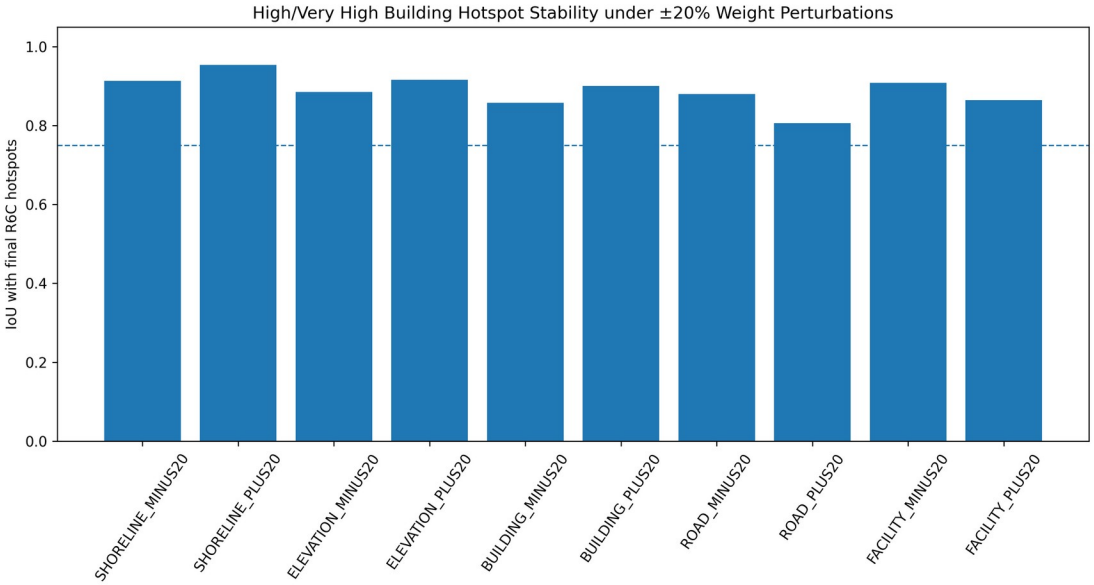
Against the historical impact-band proxy, the final classification achieved 92.74% recall, 93.45% precision and 82.66% specificity. The continuous CVI achieved ROC-AUC 0.9531.



Historical validation performance.

6. Robustness

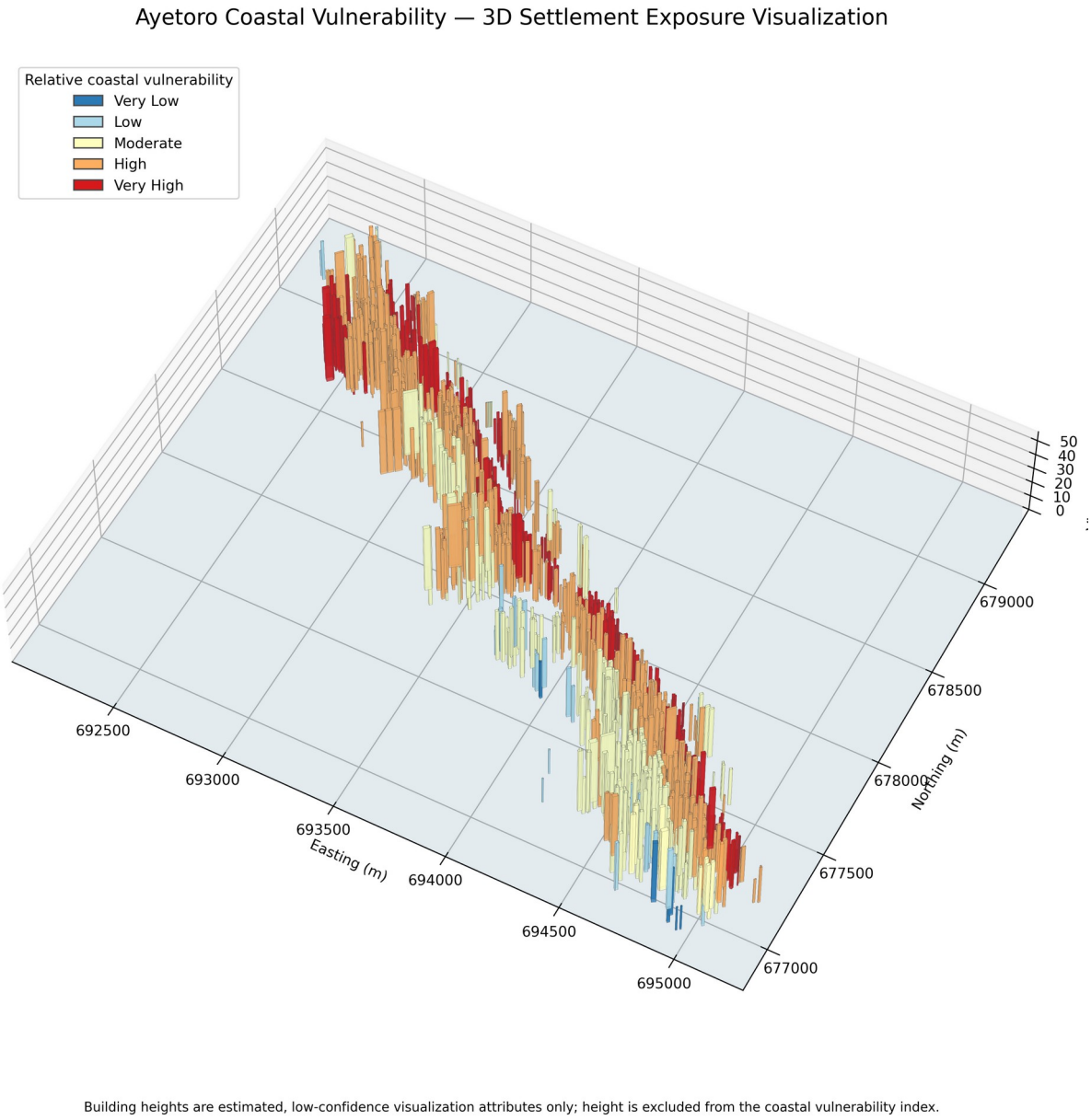
Each indicator weight was varied by +/-20% one at a time. The minimum High/Very High hotspot Intersection over Union was 0.783 for the raster result and 0.806 for building hotspots. The continuous spatial ranking remained highly stable, with minimum raster rank correlation above 0.999.



Stability of High/Very High vulnerability hotspots under formal weight perturbations.

7. 3D Planning Visualisation

The 3D component is a communication tool rather than a model input. No measured building-height dataset was available for the final inventory. All building heights are estimated from footprint-area classes, marked as low confidence and used only to extrude building footprints visually.



3D settlement exposure visualisation. Heights are estimated for communication only.

8. Planning Value

The model provides an evidence-based screening layer for prioritising detailed site investigation, infrastructure protection, preparedness and longer-term coastal adaptation planning. It can help

distinguish where building exposure is spatially concentrated before more detailed engineering, socioeconomic and community-level assessments are commissioned.

9. Limitations

The present-day building inventory is not a reconstruction of the 2018 settlement. A current building inside the historical impact band should not be interpreted automatically as a historically submerged building. The vulnerability classes are relative to Ayetoro's observed CVI distribution and are not universal coastal-hazard thresholds.

Google Open Buildings is an independent supplementation source rather than perfect ground truth. Estimated 3D heights do not represent measured morphology. The analysis does not replace coastal engineering design, household-level socioeconomic assessment, land-tenure review, environmental assessment or community consultation.

10. Citation

Abdullah Abdazeez Ayomide (2026). Coastal Vulnerability and 3D Planning Visualisation - Ayetoro, Nigeria.